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TEAM NAME: SThrive

PROJECT TITLE: SThrive Network

PROGRAMME: COMPUTER SCIENCE

COURSE: SOFTWARE ENGINEERING (DCIT 208)

LECTURER: PROFESSOR MARK ATTA-MENSAH

INDIVIDUAL GITHUB PORTFOLIO: https://github.com/Adom-Asare

GROUP GITHUB PORTFOLIO: https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-sthrive/tree/main

VIDEO URL: https://youtu.be/Aj9diOfI_Gg

DATE: 26-06-26

Typed Explanation – SThrive Network Registration Feature

1. What device is the user on?

The users of the SThrive Network platform may use any device capable of connecting to the internet and accessing a web browser, such as a desktop computer, laptop, tablet, or smartphone. Since the software being developed is a web application rather than a mobile application, users will primarily access the system through a browser interface. The frontend is planned to be implemented using React, which allows the application to be responsive across different devices.

2. Is it a browser or mobile app?

The platform is a browser-based web application and not a dedicated mobile application. Users will interact with the system through their web browsers. React will be used to create an interactive and responsive user interface that can adapt to different screen sizes.

3. What URL, route, screen, or endpoint is involved?

The URL involved in the registration process is:

shethrivesnetwork.org/register

The API endpoint responsible for handling the registration request is:

/api/auth/register

Users access the registration page through the browser, complete the required information, and submit the registration form. The request is then sent to the backend endpoint for processing.

4. What protocol is used?

The registration process will use HTTPS instead of HTTP. HTTPS encrypts information transferred between the user's browser and the web server. This prevents attackers from intercepting sensitive information such as passwords or email addresses while data is travelling across the network. Since SThrive Network is a charity organization and users will be providing personal information, ensuring secure transmission of data is important.

5. What data is sent?

The registration data will be sent from the frontend to the backend in JSON format because JSON provides a lightweight and structured way of transferring data between systems.

The JSON request body will contain:

```
{
  "name": "Jane Doe",
  "email": "jane@example.com",
  "password": "password123",
```

"subscription": "Free"

}

The subscription field indicates the type of account selected by the user, which may be Free, Pro, or Elite.

6. What validates the request?

Both frontend and backend validation will be used.

Frontend validation will immediately check:

- Required fields
- Email format
- Password rules
- Subscription selection validity

This provides immediate feedback to users before the request reaches the server.

Backend validation performs additional checks such as:

- Rechecking email format and password requirements
- Verifying whether the email already exists
- Confirming that all required fields are valid
- Ensuring the selected subscription value is one of the permitted values (Free, Pro, or Elite)

Backend validation acts as the main security layer because frontend validation can potentially be bypassed.

7. What database table, collection, or external service is touched?

The application currently plans to use MongoDB Atlas as the database service. User information will be stored inside a Users collection.

The fields stored may include:

- User ID
- Name
- Email
- Password hash
- Subscription type

The email field will have a unique index so that no two users can register with the same email address. Redis is also planned as an external service for implementing rate limiting to control excessive requests.

8. What response comes back?

The response depends on whether the registration request succeeds or fails.

Successful registration:

Status Code: 201 Created

{

"success": true,

"id":"64f1a2b9",

"email":"jane@example.com"

}

Failed registration due to duplicate email:

Status Code: 409 Conflict

{

"success": false,

"error":"Email already exists"

}

Unexpected server problem:

Status Code: 500 Internal Server Error

{

"success": false,

"error":"Internal server error"

}

9. What happens if validation fails?

If validation fails, the registration request will not proceed successfully. The user interface will display an error message showing the reason for the failure. The relevant input field will be highlighted in red while the page itself remains loaded. This approach provides immediate feedback without forcing users to refresh or restart the registration process.

10. What happens if the database is down?

If MongoDB Atlas becomes unavailable, the system should return a graceful failure response instead of appearing unresponsive. A status code of 503 Service Unavailable should be returned because the problem results from a service issue rather than a client issue.

Example:

{

"success": false,

"error":"Service unavailable. Please try again later."

}

This prevents the application from hanging indefinitely and provides users with understandable feedback.

11. What happens if 500 users do this at once?

If many users attempt registration simultaneously, the server may experience heavy load. Duplicate registration race conditions could occur if multiple users attempt to register with the same email at almost the same time. The email unique index in MongoDB helps prevent duplicate account creation.

Redis rate limiting will also be implemented. Redis will monitor requests originating from individual IP addresses and restrict excessive requests before they reach database logic. This reduces unnecessary server load and helps protect the system against spam requests or abuse.

If the system becomes overloaded, it should fail gracefully and return:

Status Code: 503 Service Unavailable

rather than hanging or crashing completely.

12. What security or privacy concern exists?

Several security and privacy concerns exist within web applications.

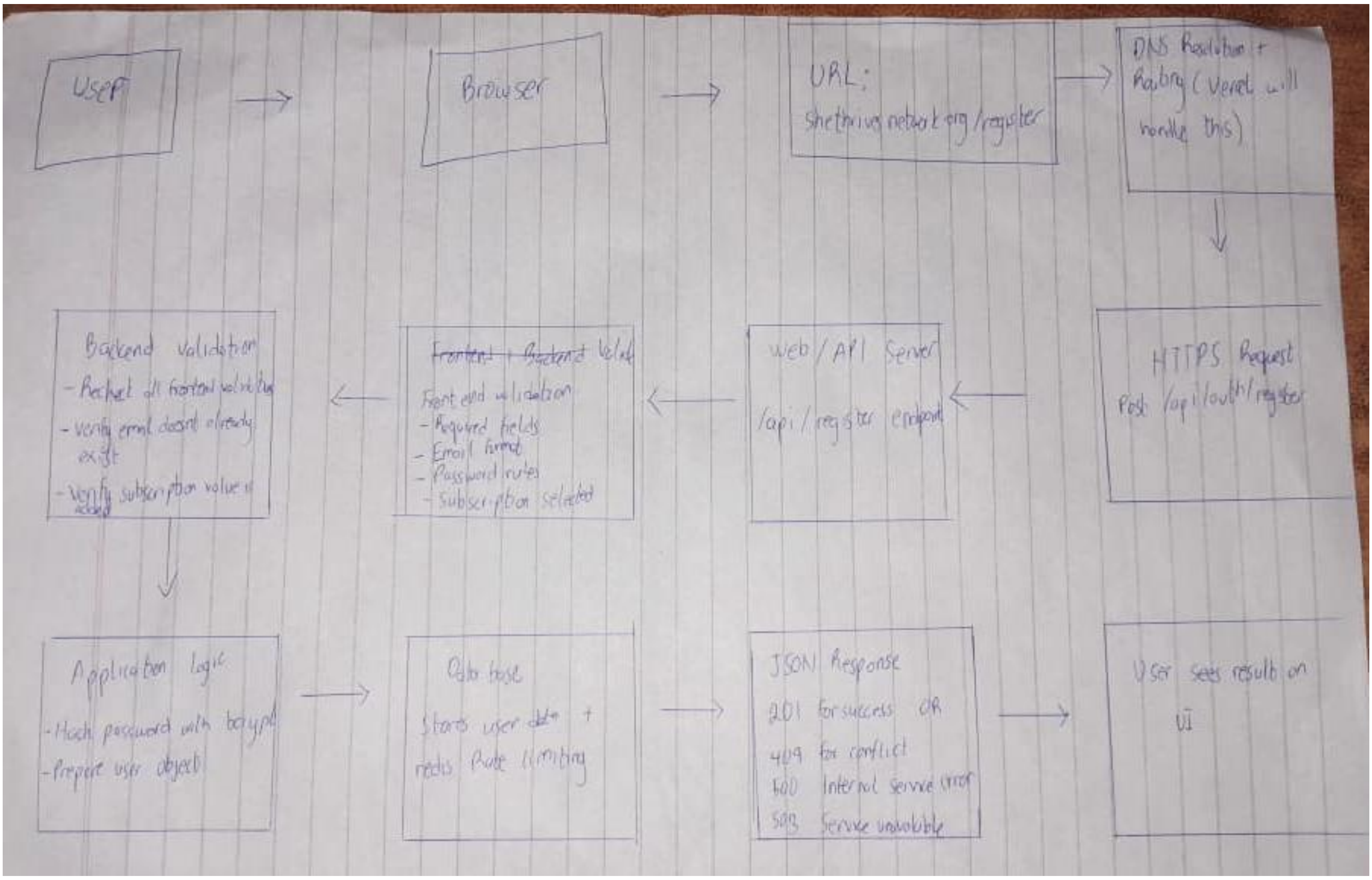
One concern is database breaches. If passwords are stored in plain text, attackers could gain access to user passwords if the database is compromised. To reduce this risk, bcrypt password hashing will be used. Hashing is a one-way process that prevents original passwords from being recovered.

Another concern involves data interception during transmission. Without HTTPS, attackers on unsecured networks could intercept user information while it travels between the browser and server. HTTPS reduces this risk by encrypting transmitted data.

Another possible risk is malicious user input. Attackers could attempt SQL injection or cross-site scripting attacks by entering harmful input. Input validation and sanitization help prevent these attacks.

Finally, privacy concerns also exist because SThrive Network is a UK-based charity organization and UK GDPR regulations apply. Collecting unnecessary personal information or processing user data without consent can create legal and ethical issues. Therefore, the system should only collect necessary information and obtain proper user consent before processing data.

Flow Diagram



AI Disclosure Statement

I used AI assistance. The tool used was ChatGPT. The purpose of use was to improve clarity, organization, grammar, and formatting of my independently written request-flow explanation, as well as to help refine technical explanations and verify system design concepts. The AI was used to reorganize ideas into a clearer structure and improve readability while preserving the original project-specific content and logic.

Date	Tool	Prompt Summary	Output Used	Output Rejected	Verification Method
June 2026	ChatGPT	Improve structure, grammar, and organization of request-flow explanation and verify technical concepts	Improved wording, organization, formatting, and clearer explanations	Generic assumptions unrelated to the project and any inaccurate technical suggestions	Compared with project requirements, lecture materials, and team project decisions before final inclusion

I understand that I must be able to explain and defend every part of this submission.